

Online Appendix for ‘Macroeconomic Stabilization and Endogenous Inequality in a TANK Model with Dual Nominal Rigidities’

This online appendix reports the nonlinear primitives of the model and the intermediate log-linearization steps underlying Section 3 of the main text. Appendix A in the main text derives the analytical reduction from the nonredundant 20-equation system to the five-equation canonical representation.

Model Description

The appendix first states the nonlinear problems of the labor packer, households, firms, and government. It then collects a reduced nonlinear equilibrium system sufficient for the first-order approximation, and finally reports the nonredundant log-linear system used in Section 3 of the main text.

Table 1 summarizes the main variables and objects that appear before the log-linear approximation.

Table 1: Nonlinear variables and objects introduced before log-linearization

Category	Notation	Meaning
Indices	$h \in [0, 1]$	Household or labor-variety index
Indices	$j \in [0, 1]$	Intermediate-good firm or goods-variety index
Types	$i \in \{H, S\}$	Household type: hand-to-mouth (H) or saver (S)
Prices and wages	P_t	Final-good price index
Prices and wages	$P_t(j)$	Price set by intermediate-good firm j
Prices and wages	$W_t^N, W_t^N(h)$	Aggregate nominal wage and household-specific nominal wage

Category	Notation	Meaning
Prices and wages	$W_t^{N,i}$	Type- i nominal wage
Prices and wages	$W_t \equiv W_t^N/P_t$	Aggregate real wage
Prices and wages	$W_t^i \equiv W_t^{N,i}/P_t$	Type- i real wage
Inflation rates	$\Pi_t^p \equiv P_t/P_{t-1}$	Gross price inflation
Inflation rates	$\Pi_t^i \equiv W_t^{N,i}/W_{t-1}^{N,i}$	Type- i gross wage inflation
Inflation rates	Π_t^w	Gross inflation of the aggregate wage index
Quantities	$C_t, C_t(h), C_t^i$	Aggregate, household-specific, and type- i consumption
Quantities	$N_t, N_t(h), N_t^i$	Aggregate, household-specific, and type- i labor
Quantities	$Y_t, Y_t(j)$	Aggregate final output and output of intermediate-good firm j
Preferences	$U(C, N)$	Period utility function
Preferences	$U_C(C, N), U_N(C, N)$	Marginal utilities of consumption and labor
Preferences	$MRS(C, N) \equiv -U_N(C, N)/U_C(C, N)$	Marginal rate of substitution
Technology	$A_t \equiv \exp(a_t)$	Aggregate productivity
Technology	$F(N), F_N(N)$	Production function and marginal product component
Technology	$MPN_t \equiv A_t F_N(N_t)$	Marginal product of labor
Markups	$MU_t^i \equiv W_t^i/MRS(C_t^i, N_t^i)$	Type- i wage markup
Markups	$MU_t^w \equiv W_t/MRS(C_t, N_t)$	Aggregate wage markup
Markups	$MU_t^p \equiv MPN_t/W_t$	Price markup
Assets and policy	$B_t^N(h), B_t^N$	Household and Type-S per-capita nominal bond holdings
Assets and policy	$V_t, \Omega_t(h)$	Ex-dividend equity value and household equity holdings
Assets and policy	$Q_{t,t+i}$	Saver stochastic discount factor from t to $t+i$
Assets and policy	R_t^N, R_t	Gross nominal interest rate and ex ante gross real interest rate
Fiscal variables	T_t^H, T_t^S, T_t^M	Net fiscal liabilities of Type H households, Type S households, and firms
Fiscal variables	Z_t, Z	Transfer to Type H households and its steady-state component
Fiscal variables	G_t, G	Government purchases and their steady-state component
Firms	$D_t(j), D_t$	Firm- j and aggregate real dividends

Category	Notation	Meaning
Shocks	a_t, m_t, z_t, g_t	Productivity, monetary-policy, redistribution, and government-spending shocks
Innovations	$e_t^a, e_t^m, e_t^z, e_t^g$	Innovations to the exogenous processes
Multipliers	$\Lambda_t^L, \Lambda_t^F, \Lambda_t^M(j)$	Multipliers for the labor packer, final-goods firm, and intermediate firm
Multipliers	$\Lambda_t^H(h), \Lambda_t^S(h)$	Type-H and Type-S household budget multipliers

Throughout, upper-case letters denote level variables. The index $h \in [0, 1]$ refers to households, and the index $j \in [0, 1]$ refers to intermediate-goods firms. A superscript $i \in \{H, S\}$ indicates the household type, where (H) denotes hand-to-mouth households and S denotes saver households. P_t denotes the final-good price index, W_t^N the aggregate nominal wage, and $W_t \equiv W_t^N/P_t$ the aggregate real wage. For each household type i , C_t^i , N_t^i , and $W_t^{N,i}$ denote consumption, labor supply, and the nominal wage, while $W_t^i \equiv W_t^{N,i}/P_t$ denotes the corresponding real wage. Aggregate price inflation is defined as $\Pi_t^p \equiv P_t/P_{t-1}$, and type-specific wage inflation is defined as $\Pi_t^i \equiv W_t^{N,i}/W_{t-1}^{N,i}$. Preferences are represented by a general period utility function $U(C, N)$, with marginal utilities $U_C(C, N) > 0$ and $U_N(C, N) < 0$. The marginal rate of substitution is $MRS(C, N) \equiv -U_N(C, N)/U_C(C, N)$. Intermediate firms use the general technology $Y = A_t F(N)$, with $F_N(N) > 0$. The marginal product of labor is $MPN_t \equiv A_t F_N(N_t)$. The type- i wage markup is $MU_t^i \equiv W_t^i/MRS(C_t^i, N_t^i)$, and the price markup is $MU_t^p \equiv MPN_t/W_t$. Finally, R_t^N denotes the gross nominal interest rate and D_t firm dividends. The ex-dividend value of the aggregate equity claim is V_t , and $\Omega_t(h)$ denotes household h 's equity holdings. Saver households price claims with $Q_{t,t+i} \equiv \beta^i U_C(C_{t+i}^S, N_{t+i}^S)/U_C(C_t^S, N_t^S)$. Here T_t^i denotes the net fiscal liability of type- i households. A negative value therefore corresponds to a transfer received by that household type. The variable Z_t denotes the transfer directed to Type H households. Government purchases are denoted by G_t , and $\Gamma \equiv G/Y$ is the steady-state government-spending share. In the redistribution rule considered below, the steady-state transfer is chosen so that type-specific steady-state consumption is equalized, $C^H = C^S$.

Labor Packer

The labor packer is a competitive aggregator that combines differentiated household labor into a homogeneous labor input sold to intermediate-goods firms. Household h supplies labor $N_t(h)$ and is paid the nominal wage $W_t^N(h)$, while the packer sells the aggregate labor input N_t at the aggregate nominal wage W_t^N . The parameter $\psi_w > 1$ is the elasticity of substitution across labor varieties, and Λ_t^L is the multiplier on the constant-elasticity-of-substitution aggregation constraint. Taking the household-specific wages $\{W_t^N(h)\}_{h \in [0,1]}$ as given, the labor packer chooses $\{N_t(h)\}_{h \in [0,1]}$ and N_t , which yields the labor demand schedule for each household and the aggregate nominal wage index.

- Objects taken as given: $P_t, W_t^N, \{W_t^N(h)\}_{h \in [0,1]}$
- Choice variables: $N_t, \{N_t(h)\}_{h \in [0,1]}$
- Objective Function:

$$W_t^N N_t - \int_0^1 W_t^N(h) N_t(h) dh$$

- Constraints:

$$N_t = \left(\int_0^1 N_t(h)^{\frac{\psi_w - 1}{\psi_w}} dh \right)^{\frac{\psi_w}{\psi_w - 1}}, \quad \psi_w > 1.$$

- Lagrangian:

$$\begin{aligned} \mathcal{L}_t = & W_t^N N_t - \int_0^1 W_t^N(h) N_t(h) dh \\ & + \Lambda_t^L \left(\left(\int_0^1 N_t(h)^{\frac{\psi_w - 1}{\psi_w}} dh \right)^{\frac{\psi_w}{\psi_w - 1}} - N_t \right) \end{aligned}$$

- FOC wrt N_t :

$$W_t^N = \Lambda_t^L$$

- FOC wrt $N_t(h)$:

$$N_t(h) = N_t \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w}$$

- Nominal wage:

$$W_t^N = \left[\int_0^1 W_t^N(h)^{1-\psi_w} dh \right]^{\frac{1}{1-\psi_w}}$$

Households

There is a continuum of households indexed by $h \in [0, 1]$. Each household consumes $C_t(h)$, supplies differentiated labor $N_t(h)$, and sets its own nominal wage $W_t^N(h)$. Preferences are common across households and are represented by period utility $U(C_t(h), N_t(h))$. Wage setting is subject to Rotemberg-type adjustment costs governed by η_w , and labor income is subsidized at rate τ_w . A fraction λ of households are hand-to-mouth households (Type H), who do not trade assets, while the remaining fraction $1 - \lambda$ are saver households (Type S), who hold one-period nominal bonds and receive firm dividends. Within each type, full risk sharing implies identical consumption, labor, and wage choices in equilibrium.

With two household types and within-type symmetry, the labor-packer aggregation functions defined and derived in the previous subsection can be rewritten as follows:

$$\begin{aligned}
N_t &= \left[\lambda (N_t^H)^{\frac{\psi_w-1}{\psi_w}} + (1-\lambda) (N_t^S)^{\frac{\psi_w-1}{\psi_w}} \right]^{\frac{\psi_w}{\psi_w-1}} \\
W_t^N &= \left[\lambda (W_t^{N,H})^{1-\psi_w} + (1-\lambda) (W_t^{N,S})^{1-\psi_w} \right]^{\frac{1}{1-\psi_w}} \\
&= P_t \left[\lambda (W_t^H)^{1-\psi_w} + (1-\lambda) (W_t^S)^{1-\psi_w} \right]^{\frac{1}{1-\psi_w}}
\end{aligned}$$

Type H

A Type H household chooses current consumption $C_t(h)$, labor supply $N_t(h)$, and its own nominal wage $W_t^N(h)$, taking aggregate variables as given. Since it does not trade financial assets, its current uses of funds consist of consumption and the net fiscal liability T_t^H , where $T_t^H < 0$ corresponds to a transfer received from the government. In the budget constraint, P_t is the aggregate price level, Y_t is aggregate output, and η_w measures the cost of changing the nominal wage relative to the previous period's wage $W_{t-1}^N(h)$. The optimality conditions imply the marginal utility of consumption, the household-specific labor demand schedule, and the Type H wage-setting equation. In symmetric equilibrium, these conditions can be written in terms of the Type H real wage W_t^H , wage inflation Π_t^H , marginal rate of substitution $MRS_t^H \equiv MRS(C_t^H, N_t^H)$, and wage markup $MU_t^H \equiv W_t^H / MRS_t^H$.

- Objects taken as given: $W_t^N, P_t, N_t, Y_t, T_t^H, W_{t-1}^N(h)$
- Choice variables: $C_t(h), N_t(h), W_t^N(h)$
- Objective Function:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t(h), N_t(h))$$

- Constraints:

$$\begin{aligned}
&C_t(h) + T_t^H \\
&= (1 + \tau_w) \frac{W_t^N(h)}{P_t} N_t(h) - \frac{\eta_w}{2} \left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1 \right)^2 Y_t \\
&N_t(h) = N_t \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w}
\end{aligned}$$

- Lagrangian:

$$\begin{aligned}\mathbb{L} &= \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mathcal{L}_t \\ \mathcal{L}_t &= U \left(C_t(h), N_t \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w} \right) \\ &\quad + \Lambda_t^H(h) \left((1 + \tau_w) \frac{N_t}{P_t} W_t^N \left(\frac{W_t^N(h)}{W_t^N} \right)^{1-\psi_w} - \frac{\eta_w}{2} \left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1 \right)^2 Y_t \right. \\ &\quad \left. - C_t(h) - T_t^H \right)\end{aligned}$$

- FOC wrt $C_t(h)$:

$$\Lambda_t^H(h) = U_C(C_t(h), N_t(h))$$

- FOC wrt $W_t^N(h)$:

$$\begin{aligned}& -\psi_w \frac{N_t}{W_t^N} \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w-1} U_N(C_t(h), N_t(h)) \\ & + (1 - \psi_w) \Lambda_t^H(h) (1 + \tau_w) \frac{N_t}{P_t} \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w} \\ & - \Lambda_t^H(h) \eta_w \left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1 \right) \frac{Y_t}{W_{t-1}^N(h)} \\ & + \beta \mathbb{E}_t \Lambda_{t+1}^H(h) \eta_w \left(\frac{W_{t+1}^N(h)}{W_t^N(h)} - 1 \right) \frac{W_{t+1}^N(h) Y_{t+1}}{W_t^N(h)^2} = 0\end{aligned}$$

- Symmetric Equilibrium ($\Lambda_t^H(h) = \Lambda_t^H$, $C_t(h) = C_t^H$, $N_t(h) = N_t^H$, $W_t^N(h) = W_t^{N,H}$):

$$\begin{aligned}N_t^H &= N_t \left(\frac{W_t^{N,H}}{W_t^N} \right)^{-\psi_w} \\ \Lambda_t^H &= U_C(C_t^H, N_t^H)\end{aligned}$$

$$\begin{aligned}\eta_w \frac{Y_t}{N_t^H} \frac{W_t^{N,H}}{W_{t-1}^{N,H}} \left(\frac{W_t^{N,H}}{W_{t-1}^{N,H}} - 1 \right) &= \psi_w \left[MRS_t^H - (1 + \tau_w) \left(1 - \frac{1}{\psi_w} \right) \frac{W_t^{N,H}}{P_t} \right] \\ &\quad + \beta \eta_w \mathbb{E}_t \left[\frac{U_C(C_{t+1}^H, N_{t+1}^H)}{U_C(C_t^H, N_t^H)} \frac{Y_{t+1}}{N_t^H} \frac{W_{t+1}^{N,H}}{W_t^{N,H}} \left(\frac{W_{t+1}^{N,H}}{W_t^{N,H}} - 1 \right) \right] \\ \Leftrightarrow \eta_w \frac{W_t^{N,H}}{W_{t-1}^{N,H}} \left(\frac{W_t^{N,H}}{W_{t-1}^{N,H}} - 1 \right) &= \psi_w \frac{W_t^{N,H} N_t^H}{P_t Y_t} \left[\frac{1}{M U_t^H} - (1 + \tau_w) \left(1 - \frac{1}{\psi_w} \right) \right] \\ &\quad + \beta \eta_w \mathbb{E}_t \left[\frac{U_C(C_{t+1}^H, N_{t+1}^H)}{U_C(C_t^H, N_t^H)} \frac{Y_{t+1}}{Y_t} \frac{W_{t+1}^{N,H}}{W_t^{N,H}} \left(\frac{W_{t+1}^{N,H}}{W_t^{N,H}} - 1 \right) \right]\end{aligned}$$

Type S

A Type S household chooses consumption $C_t(h)$, labor supply $N_t(h)$, its own nominal wage $W_t^N(h)$, one-period nominal bond holdings $B_t^N(h)$, and equity holdings $\Omega_t(h)$. In addition to current after-subsidy labor income, it receives gross nominal bond returns $R_{t-1}^N B_{t-1}^N(h)$ and the payoff on its equity holdings, $\Omega_{t-1}(h)(V_t + D_t)$, where V_t is the ex-dividend value of the aggregate equity claim. It also faces the net fiscal liability T_t^S . A positive value of T_t^S is therefore a levy, while a negative value corresponds to a transfer. Here R_t^N denotes the gross nominal interest rate on the bond purchased in period t , and $B_t^N(h)/P_t$ is the household's real bond position. The optimality conditions deliver intertemporal asset-pricing equations and a wage-setting equation analogous to that of Type H households. In symmetric equilibrium, equity-market clearing gives $(1-\lambda)\Omega_t = 1$, so Type S households receive the per-capita dividend $D_t/(1-\lambda)$.

- Objects taken as given: $W_t^N, P_t, R_t^N, V_t, D_t, N_t, Y_t, T_t^S, W_{t-1}^N(h), B_{t-1}^N(h), \Omega_{t-1}(h)$
- Choice variables: $C_t(h), N_t(h), W_t^N(h), B_t^N(h), \Omega_t(h)$
- Objective Function:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t(h), N_t(h))$$

- Constraints:

$$\begin{aligned} C_t(h) + \frac{B_t^N(h)}{P_t} + \Omega_t(h)V_t + T_t^S \\ = (1 + \tau_w) \frac{W_t^N(h)}{P_t} N_t(h) - \frac{\eta_w}{2} \left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1 \right)^2 Y_t \\ + R_{t-1}^N \frac{B_{t-1}^N(h)}{P_t} + \Omega_{t-1}(h)(V_t + D_t) \\ N_t(h) = N_t \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w} \end{aligned}$$

- Lagrangian:

$$\begin{aligned} \mathbb{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mathcal{L}_t \\ \mathcal{L}_t = U \left(C_t(h), N_t \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w} \right) \\ + \Lambda_t^S(h) \left(\begin{aligned} & (1 + \tau_w) \frac{W_t^N N_t}{P_t} \left(\frac{W_t^N(h)}{W_t^N} \right)^{1-\psi_w} - \frac{\eta_w}{2} \left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1 \right)^2 Y_t \\ & + R_{t-1}^N \frac{B_{t-1}^N(h)}{P_t} + \Omega_{t-1}(h)(V_t + D_t) - C_t(h) - \frac{B_t^N(h)}{P_t} - \Omega_t(h)V_t - T_t^S \end{aligned} \right) \end{aligned}$$

- FOC wrt $C_t(h)$:

$$\Lambda_t^S(h) = U_C(C_t(h), N_t(h))$$

Define the saver stochastic discount factor from t to $t+i$ as

$$Q_{t,t+i} \equiv \beta^i \frac{U_C(C_{t+i}^S, N_{t+i}^S)}{U_C(C_t^S, N_t^S)}.$$

- FOC wrt $W_t^N(h)$:

$$\begin{aligned} & -\psi_w \frac{N_t}{W_t^N} \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w - 1} U_N(C_t(h), N_t(h)) \\ & + (1 - \psi_w) \Lambda_t^S(h) (1 + \tau_w) \frac{N_t}{P_t} \left(\frac{W_t^N(h)}{W_t^N} \right)^{-\psi_w} \\ & - \Lambda_t^S(h) \eta_w \left(\frac{W_t^N(h)}{W_{t-1}^N(h)} - 1 \right) \frac{Y_t}{W_{t-1}^N(h)} \\ & + \Lambda_t^S(h) \mathbb{E}_t \left[Q_{t,t+1} \eta_w \left(\frac{W_{t+1}^N(h)}{W_t^N(h)} - 1 \right) \frac{W_{t+1}^N(h) Y_{t+1}}{W_t^N(h)^2} \right] = 0 \end{aligned}$$

- FOC wrt $B_t^N(h)$:

$$\frac{1}{P_t} = \mathbb{E}_t \left[Q_{t,t+1} \frac{R_t^N}{P_{t+1}} \right]$$

- FOC wrt $\Omega_t(h)$:

$$V_t = \mathbb{E}_t [Q_{t,t+1} (V_{t+1} + D_{t+1})]$$

- Symmetric Equilibrium ($\Lambda_t^S(h) = \Lambda_t^S$, $C_t(h) = C_t^S$, $N_t(h) = N_t^S$, $W_t^N(h) = W_t^{N,S}$, where W_t^S is the real wage of type S):

$$\begin{aligned} N_t^S &= N_t \left(\frac{W_t^{N,S}}{W_t^N} \right)^{-\psi_w} \\ \Lambda_t^S &= U_C(C_t^S, N_t^S) \\ 1 &= \mathbb{E}_t \left[Q_{t,t+1} R_t^N \frac{P_t}{P_{t+1}} \right] \\ V_t &= \mathbb{E}_t [Q_{t,t+1} (V_{t+1} + D_{t+1})] \\ &= \mathbb{E}_t \sum_{i=1}^{\infty} Q_{t,t+i} D_{t+i} \end{aligned}$$

$$\begin{aligned}
\eta_w \frac{Y_t}{N_t^S} \frac{W_t^{N,S}}{W_{t-1}^{N,S}} \left(\frac{W_t^{N,S}}{W_{t-1}^{N,S}} - 1 \right) &= \psi_w \left[MRS_t^S - (1 + \tau_w) \left(1 - \frac{1}{\psi_w} \right) \frac{W_t^{N,S}}{P_t} \right] \\
&\quad + \eta_w \mathbb{E}_t \left[Q_{t,t+1} \frac{Y_{t+1}}{N_t^S} \frac{W_{t+1}^{N,S}}{W_t^{N,S}} \left(\frac{W_{t+1}^{N,S}}{W_t^{N,S}} - 1 \right) \right] \\
\Leftrightarrow \eta_w \frac{W_t^{N,S}}{W_{t-1}^{N,S}} \left(\frac{W_t^{N,S}}{W_{t-1}^{N,S}} - 1 \right) &= \psi_w \frac{W_t^{N,S} N_t^S}{P_t Y_t} \left[\frac{1}{MU_t^S} - (1 + \tau_w) \left(1 - \frac{1}{\psi_w} \right) \right] \\
&\quad + \eta_w \mathbb{E}_t \left[Q_{t,t+1} \frac{Y_{t+1}}{Y_t} \frac{W_{t+1}^{N,S}}{W_t^{N,S}} \left(\frac{W_{t+1}^{N,S}}{W_t^{N,S}} - 1 \right) \right]
\end{aligned}$$

where $MRS_t^S = MRS(C_t^S, N_t^S)$ is Type S marginal rate of substitution, and $MU_t^S = W_t^S / MRS_t^S$ is Type S wage markup.

Final Goods Firm

The final-goods firm is perfectly competitive and combines differentiated intermediate goods into a homogeneous final good using a constant-elasticity-of-substitution aggregator. The quantity of variety j is denoted by $Y_t(j)$, its price by $P_t(j)$, aggregate final output by Y_t , and the final-good price index by P_t . The elasticity of substitution across intermediate varieties is $\psi_p \geq 1$, and Λ_t^F is the multiplier on the aggregation constraint. Taking the prices $\{P_t(j)\}_{j \in [0,1]}$ as given, the final-goods firm chooses $(\{Y_t(j)\}_{j \in [0,1]})$ and Y_t , which yields the demand schedule for each intermediate-good producer and the aggregate final-good price index.

- Objects taken as given: $P_t, \{P_t(j)\}_{j \in [0,1]}$
- Choice variables: $Y_t, \{Y_t(j)\}_{j \in [0,1]}$
- Objective Function:

$$P_t Y_t - \int_0^1 P_t(j) Y_t(j) dj$$

- Constraints:

$$Y_t = \left(\int_0^1 Y_t(j)^{\frac{\psi_p - 1}{\psi_p}} dj \right)^{\frac{\psi_p}{\psi_p - 1}}, \quad \psi_p \geq 1$$

- Lagrangian:

$$\begin{aligned}
\mathcal{L}_t &= P_t Y_t - \int_0^1 P_t(j) Y_t(j) dj \\
&\quad + \Lambda_t^F \left(\left(\int_0^1 Y_t(j)^{\frac{\psi_p - 1}{\psi_p}} dj \right)^{\frac{\psi_p}{\psi_p - 1}} - Y_t \right)
\end{aligned}$$

- FOC wrt Y_t :

$$\Lambda_t^F = P_t$$

- FOC wrt $Y_t(j)$:

$$Y_t(j) = Y_t \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p}$$

- Price of Final Goods:

$$P_t = \left[\int_0^1 P_t(j)^{1-\psi_p} dj \right]^{\frac{1}{1-\psi_p}}$$

Intermediate Goods Firms

Intermediate-good firm j produces output $Y_t(j)$ using labor $N_t(j)$ according to the general technology $Y_t(j) = A_t F(N_t(j))$, where $A_t \equiv \exp(a_t)$ denotes aggregate labor productivity and $F_N(N) > 0$. The firm hires homogeneous labor from the labor packer at the nominal wage W_t^N , sets its own price $P_t(j)$, and faces Rotemberg price adjustment costs governed by η_p . Its real profits are denoted by $D_t(j)$, the government levies the lump-sum tax T_t^M , and the sales subsidy rate is τ_p . The firm's objective is the date-0 present discounted value of profits, discounted by the saver household's stochastic discount factor $Q_{0,t}$. The optimality conditions determine labor demand, real marginal cost, and the price-setting equation. In symmetric equilibrium, the price markup is $MU_t^p \equiv P_t MPN_t / W_t^N = MPN_t / W_t$ and $MPN_t \equiv A_t F_N(N_t)$ is the marginal product of labor.

- Objects taken as given: $W_t^N, P_t, Y_t, A_t, T_t^M, C_t^S, P_{t-1}(j)$
- Choice variables: $N_t(j), P_t(j), Y_t(j)$
- Objective Function:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} Q_{0,t} D_t(j)$$

- Constraints:

$$\begin{aligned} D_t(j) &= (1 + \tau_p) \frac{P_t(j)}{P_t} Y_t(j) - \frac{\eta_p}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - 1 \right)^2 Y_t - \frac{W_t^N}{P_t} N_t(j) - T_t^M \\ Y_t(j) &= Y_t \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p} \\ Y_t(j) &= A_t F(N_t(j)) \end{aligned}$$

- Lagrangian:

$$\begin{aligned}\mathbb{L} = & \mathbb{E}_0 \sum_{t=0}^{\infty} Q_{0,t} \mathcal{L}_t \\ \mathcal{L}_t = & (1 + \tau_p) Y_t \left(\frac{P_t(j)}{P_t} \right)^{1-\psi_p} - \frac{\eta_p}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - 1 \right)^2 Y_t - \frac{W_t^N}{P_t} N_t(j) \\ & - T_t^M + \Lambda_t^M(j) \left(A_t F(N_t(j)) - Y_t \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p} \right)\end{aligned}$$

- FOC wrt $N_t(j)$:

$$\Lambda_t^M(j) A_t F_N(N_t(j)) = W_t^N / P_t$$

- FOC wrt $P_t(j)$:

$$\begin{aligned}& Q_{0,t} (1 + \tau_p) (1 - \psi_p) \frac{Y_t}{P_t} \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p} - Q_{0,t} \eta_p \frac{Y_t}{P_{t-1}(j)} \left(\frac{P_t(j)}{P_{t-1}(j)} - 1 \right) \\ & + Q_{0,t} \psi_p \Lambda_t^M(j) \frac{Y_t}{P_t} \left(\frac{P_t(j)}{P_t} \right)^{-\psi_p - 1} \\ & + \eta_p \mathbb{E}_t \left[Q_{0,t+1} Y_{t+1} \frac{P_{t+1}(j)}{P_t(j)^2} \left(\frac{P_{t+1}(j)}{P_t(j)} - 1 \right) \right] = 0\end{aligned}$$

- Symmetric Equilibrium ($N_t(j) = N_t$, $\Lambda_t^M(j) = \Lambda_t^M$, $P_t(j) = P_t$, $Y_t(j) = Y_t$):

$$\begin{aligned}\eta_p \frac{P_t}{P_{t-1}} \left(\frac{P_t}{P_{t-1}} - 1 \right) = & \psi_p \left[\frac{W_t^N / P_t}{M P N_t} - (1 + \tau_p) \left(1 - \frac{1}{\psi_p} \right) \right] \\ & + \eta_p \mathbb{E}_t \left[Q_{t,t+1} \frac{Y_{t+1}}{Y_t} \frac{P_{t+1}}{P_t} \left(\frac{P_{t+1}}{P_t} - 1 \right) \right] \\ = & \psi_p \left[\frac{1}{M U_t^p} - (1 + \tau_p) \left(1 - \frac{1}{\psi_p} \right) \right] \\ & + \eta_p \mathbb{E}_t \left[Q_{t,t+1} \frac{Y_{t+1}}{Y_t} \frac{P_{t+1}}{P_t} \left(\frac{P_{t+1}}{P_t} - 1 \right) \right]\end{aligned}$$

- Profits:

$$D_t = (1 + \tau_p) Y_t - \frac{\eta_p}{2} \left(\frac{P_t}{P_{t-1}} - 1 \right)^2 Y_t - \frac{W_t^N}{P_t} N_t - T_t^M$$

When $(1 + \tau_p)(1 - 1/\psi_p) = 1$, steady state price markup is one.

Government

The government performs three functions in the model. First, it provides the wage subsidy τ_w and the sales subsidy τ_p , eliminating steady-state markups under the baseline specialization introduced below. Second, it redistributes resources across household types through the lump-sum components T_t^H and T_t^S , and finances government purchases G_t with type-invariant lump-sum liabilities. Here Z_t denotes the transfer received by Type H households, while Type S households finance that transfer through a balanced-budget levy. Because T_t^i denotes net fiscal liability, the transfer implies $T_t^H < 0$ for the recipient household type and a positive levy on Type S through T_t^S . The steady-state component of the transfer is chosen to equalize type-specific consumption in the steady state. Third, the monetary authority sets the gross nominal interest rate R_t^N according to a Taylor rule that responds to gross price inflation Π_t^p and to the monetary policy shock m_t .

Fiscal policy:

$$\begin{aligned}
T_t^H &= -Z_t + \tau_w W_t^H N_t^H + G_t \\
T_t^S &= \frac{\lambda}{1-\lambda} Z_t + \tau_w W_t^S N_t^S + G_t \\
T_t^M &= \tau_p Y_t \\
\tau_w &= 1/(\psi_w - 1) \\
\tau_p &= 1/(\psi_p - 1) \\
Z_t &= Z + z_t Y \\
Z &= D \\
C^H &= C^S \\
z_t &= \rho_z z_{t-1} + e_t^z \\
G_t &= G + g_t Y \\
\Gamma &= G/Y \\
g_t &= \rho_g g_{t-1} + e_t^g
\end{aligned}$$

Monetary Policy:

$$\begin{aligned}
R_t^N &= (1/\beta) \exp(-m_t) (\Pi_t^p)^\phi \\
m_t &= \rho_m m_{t-1} + e_t^m
\end{aligned}$$

Equilibrium conditions

Equilibrium requires market clearing in goods, bond, and equity markets, together with consistency between aggregate variables and type-specific allocations. Let B_t^N denote Type-S per-capita nominal bond holdings in symmetric equilibrium. Bond-market clearing requires zero net bond supply, so $(1-\lambda)B_t^N = 0$. Equity supply is normalized to one, so equity-market

clearing requires $(1 - \lambda)\Omega_t = 1$. Aggregate consumption C_t is the population-weighted average of C_t^H and C_t^S , and final output Y_t is allocated to private consumption, government purchases, and price and wage adjustment costs.

Final goods market:

$$Y_t = C_t + G_t + \frac{\eta_p}{2} \left(\frac{P_t}{P_{t-1}} - 1 \right)^2 Y_t + \lambda \frac{\eta_w}{2} \left(\frac{W_t^{N,H}}{W_{t-1}^{N,H}} - 1 \right)^2 Y_t + (1 - \lambda) \frac{\eta_w}{2} \left(\frac{W_t^{N,S}}{W_{t-1}^{N,S}} - 1 \right)^2 Y_t$$

Bond market: $(1 - \lambda)B_t^N = 0$.

Equity market: $(1 - \lambda)\Omega_t = 1$.

Aggregation condition:

$$C_t = \lambda C_t^H + (1 - \lambda) C_t^S$$

Reduced nonlinear equilibrium system

Before summarizing the model, I collect the key real variables and inflation rates used below. In particular, $W_t \equiv W_t^N/P_t$ denotes the aggregate real wage, $W_t^H \equiv W_t^{N,H}/P_t$ and $W_t^S \equiv W_t^{N,S}/P_t$ denote type-specific real wages, $\Pi_t^P \equiv P_t/P_{t-1}$ denotes gross price inflation, and Π_t^H , Π_t^S , and Π_t^w denote the corresponding gross wage inflation rates for Type H, Type S, and the aggregate wage index. The aggregate wage markup is defined as $MU_t^w \equiv W_t/MRS(C_t, N_t)$. The saver stochastic discount factor is $Q_{t,t+i} \equiv \beta^i U_C(C_{t+i}^S, N_{t+i}^S)/U_C(C_t^S, N_t^S)$.

The following block collects a reduced set of nonlinear equilibrium conditions that is sufficient for the log-linearization below.

I report only the nonlinear conditions needed to obtain the first-order system below. Conditions omitted here are either redundant or recoverable by symmetry and aggregation.

The nonlinear equilibrium conditions can be summarized as follows:

- Labor Packer:

$$W_t = [\lambda(W_t^H)^{1-\psi_w} + (1 - \lambda)(W_t^S)^{1-\psi_w}]^{\frac{1}{1-\psi_w}}$$

$$N_t = \left[\lambda(N_t^H)^{\frac{\psi_w-1}{\psi_w}} + (1 - \lambda)(N_t^S)^{\frac{\psi_w-1}{\psi_w}} \right]^{\frac{\psi_w}{\psi_w-1}}$$

- Type H:

$$\begin{aligned}
N_t^H &= N_t \left(\frac{W_t^H}{W_t} \right)^{-\psi_w} \\
C_t^H &= W_t^H N_t^H + Z_t - G_t - \frac{\eta_w}{2} (\Pi_t^H - 1)^2 Y_t \\
\eta_w \Pi_t^H (\Pi_t^H - 1) &= \psi_w \left\{ \frac{1}{MU_t^H} - 1 \right\} \frac{W_t^H N_t^H}{Y_t} \\
&\quad + \eta_w \beta \mathbb{E}_t \left[\frac{U_C(C_{t+1}^H, N_{t+1}^H)}{U_C(C_t^H, N_t^H)} \frac{Y_{t+1}}{Y_t} \Pi_{t+1}^H (\Pi_{t+1}^H - 1) \right] \\
\Pi_t^H &= \frac{W_t^H}{W_{t-1}^H} \Pi_t^p \\
MU_t^H &= \frac{W_t^H}{MRS(C_t^H, N_t^H)}
\end{aligned}$$

- Type S:

$$\begin{aligned}
N_t^S &= N_t \left(\frac{W_t^S}{W_t} \right)^{-\psi_w} \\
Q_{t,t+i} &= \beta^i \frac{U_C(C_{t+i}^S, N_{t+i}^S)}{U_C(C_t^S, N_t^S)} \\
1 &= \mathbb{E}_t \left[Q_{t,t+1} \frac{R_t^N}{\Pi_{t+1}^p} \right] \\
V_t &= \mathbb{E}_t [Q_{t,t+1} (V_{t+1} + D_{t+1})] = \mathbb{E}_t \sum_{i=1}^{\infty} Q_{t,t+i} D_{t+i} \\
C_t^S &= W_t^S N_t^S + \frac{D_t}{1-\lambda} - \frac{\lambda}{1-\lambda} Z_t - G_t - \frac{\eta_w}{2} (\Pi_t^S - 1)^2 Y_t \\
\eta_w \Pi_t^S (\Pi_t^S - 1) &= \psi_w \left\{ \frac{1}{MU_t^S} - 1 \right\} \frac{W_t^S N_t^S}{Y_t} \\
&\quad + \eta_w \mathbb{E}_t \left[Q_{t,t+1} \frac{Y_{t+1}}{Y_t} \Pi_{t+1}^S (\Pi_{t+1}^S - 1) \right] \\
\Pi_t^S &= \frac{W_t^S}{W_{t-1}^S} \Pi_t^p \\
MU_t^S &= \frac{W_t^S}{MRS(C_t^S, N_t^S)}
\end{aligned}$$

- Firms:

$$\begin{aligned}
A_t &= \exp(a_t) \\
Y_t &= A_t F(N_t) \\
a_t &= \rho_a a_{t-1} + e_t^a \\
\eta_p \Pi_t^p (\Pi_t^p - 1) &= \psi_p \left\{ \frac{1}{MU_t^p} - 1 \right\} \\
&\quad + \eta_p \mathbb{E}_t \left[Q_{t,t+1} \frac{Y_{t+1}}{Y_t} \Pi_{t+1}^p (\Pi_{t+1}^p - 1) \right] \\
MU_t^p &= \frac{A_t F_N(N_t)}{W_t} \\
D_t &= Y_t - \frac{\eta_p}{2} (\Pi_t^p - 1)^2 Y_t - W_t N_t
\end{aligned}$$

- Government:

$$\begin{aligned}
R_t^N &= \frac{\exp(-m_t)}{\beta} (\Pi_t^p)^\phi \\
m_t &= \rho_m m_{t-1} + e_t^m \\
Z_t &= Z + z_t Y \\
Z &= D \\
G_t &= G + g_t Y \\
\Gamma &= G/Y \\
z_t &= \rho_z z_{t-1} + e_t^z \\
g_t &= \rho_g g_{t-1} + e_t^g
\end{aligned}$$

- Market clearing and others:

$$\begin{aligned}
Y_t &= C_t + G_t + \frac{\eta_p}{2} (\Pi_t^p - 1)^2 Y_t \\
&\quad + \lambda \frac{\eta_w}{2} (\Pi_t^H - 1)^2 Y_t + (1 - \lambda) \frac{\eta_w}{2} (\Pi_t^S - 1)^2 Y_t \\
C_t &= \lambda C_t^H + (1 - \lambda) C_t^S \\
(1 - \lambda) B_t^N &= 0 \\
(1 - \lambda) \Omega_t &= 1 \\
R_t &= \mathbb{E}_t \left[\frac{R_t^N}{\Pi_{t+1}^p} \right] \\
\Pi_t^w &= \frac{W_t}{W_{t-1}} \Pi_t^p \\
MU_t^w &= \frac{W_t}{MRS(C_t, N_t)}
\end{aligned}$$

Functional specialization and log-linear approximation

From this section onward, I specialize the primitives to the separable CRRA-Frisch utility function and set $F(N) = N^{1-\alpha}$:

$$\begin{aligned} U(C, N) &= \frac{C^{1-\gamma}}{1-\gamma} - \frac{N^{1+\varphi}}{1+\varphi}, \\ MRS(C, N) &= C^\gamma N^\varphi, \\ F(N) &= N^{1-\alpha}, \\ F_N(N) &= (1-\alpha)N^{-\alpha}. \end{aligned}$$

In the log-linearized system, lower-case letters denote log deviations from the symmetric zero-inflation steady state; for example, $c_t \equiv \log(C_t/C)$, $n_t \equiv \log(N_t/N)$, and $w_t \equiv \log(W_t/W)$. Inflation variables are log deviations of gross inflation rates. The dividend variable is the deviation of dividends from their steady-state value, scaled by steady-state output, $d_t \equiv (D_t - D)/Y$. The ex ante real interest rate is $r_t \equiv r_t^N - \mathbb{E}_t \pi_{t+1}^p$, where r_t^N is the log deviation of the gross nominal rate R_t^N . The variables a_t , m_t , $z_t \equiv (Z_t - Z)/Y$, and $g_t \equiv (G_t - G)/Y$ are exogenous deviation processes. In what follows, upper-case MU denotes level markups, whereas lower-case μ denotes their log-linear deviations. Around the symmetric zero-inflation steady state, the quadratic price- and wage-adjustment costs are second-order terms and therefore drop out from the first-order resource constraint, which implies $y_t = (1 - \Gamma)c_t + g_t$ at first order. The steady-state labor share remains $WN/Y = 1 - \alpha$.

The steady-state restrictions used below are as follows:

$$\begin{aligned} N &= N^S = N^H = \left(\frac{1-\alpha}{(1-\Gamma)^\gamma} \right)^{\frac{1}{\gamma(1-\alpha)+\varphi+\alpha}} \\ Y &= N^{1-\alpha} \\ C &= C^S = C^H = (1-\Gamma)Y \\ W &= W^S = W^H = (1-\alpha)N^{-\alpha} \\ MU^p &= MU^w = MU^S = MU^H = 1 \\ \Pi^p &= \Pi^w = \Pi^S = \Pi^H = 1 \\ D/Y &= \alpha \\ G/Y &= \Gamma \\ V/Y &= \frac{\beta}{1-\beta}\alpha \\ \Omega &= \frac{1}{1-\lambda} \end{aligned}$$

The log-linear approximation, with $d_t := (D_t - D)/Y$, is then given by:

- Labor Packer:

$$\begin{aligned}w_t &= \lambda w_t^H + (1 - \lambda) w_t^S \\n_t &= \lambda n_t^H + (1 - \lambda) n_t^S\end{aligned}$$

- Type H:

$$\begin{aligned}n_t^H &= n_t - \psi_w(w_t^H - w_t) \\c_t^H &= \frac{1 - \alpha}{1 - \Gamma}(w_t^H + n_t^H) + \frac{1}{1 - \Gamma}(z_t - g_t) \\\pi_t^H &= -(1 - \alpha)(\psi_w/\eta_w)\mu_t^H + \beta\mathbb{E}_t\pi_{t+1}^H \\\pi_t^H &= w_t^H - w_{t-1}^H + \pi_t^p \\\mu_t^H &= w_t^H - (\gamma c_t^H + \varphi n_t^H)\end{aligned}$$

- Type S:

$$\begin{aligned}n_t^S &= n_t - \psi_w(w_t^S - w_t) \\r_t^N - \mathbb{E}_t\pi_{t+1}^p &= \gamma\mathbb{E}_t\Delta c_{t+1}^S \\c_t^S &= \frac{1 - \alpha}{1 - \Gamma}(w_t^S + n_t^S) + \frac{1}{(1 - \lambda)(1 - \Gamma)}d_t - \frac{\lambda}{(1 - \lambda)(1 - \Gamma)}z_t - \frac{1}{1 - \Gamma}g_t \\\pi_t^S &= -(1 - \alpha)(\psi_w/\eta_w)\mu_t^S + \beta\mathbb{E}_t\pi_{t+1}^S \\\pi_t^S &= w_t^S - w_{t-1}^S + \pi_t^p \\\mu_t^S &= w_t^S - (\gamma c_t^S + \varphi n_t^S)\end{aligned}$$

- Firms:

$$\begin{aligned}y_t &= a_t + (1 - \alpha)n_t \\a_t &= \rho_a a_{t-1} + e_t^a \\\pi_t^p &= -(\psi_p/\eta_p)\mu_t^p + \beta\mathbb{E}_t\pi_{t+1}^p \\\mu_t^p &= y_t - n_t - w_t \\d_t &= y_t - (1 - \alpha)(w_t + n_t)\end{aligned}$$

- Government:

$$\begin{aligned}r_t^N &= \phi\pi_t^p - m_t \\m_t &= \rho_m m_{t-1} + e_t^m \\z_t &= \rho_z z_{t-1} + e_t^z \\g_t &= \rho_g g_{t-1} + e_t^g\end{aligned}$$

- Market clearing and others:

$$\begin{aligned}
y_t &= (1 - \Gamma)c_t + g_t \\
r_t &= r_t^N - \mathbb{E}_t \pi_{t+1}^p \\
c_t &= \lambda c_t^H + (1 - \lambda)c_t^S \\
\pi_t^w &= \lambda \pi_t^H + (1 - \lambda)\pi_t^S \\
\pi_t^w &= w_t - w_{t-1} + \pi_t^p \\
\pi_t^w &= -(1 - \alpha)(\psi_w/\eta_w)\mu_t^w + \beta \mathbb{E}_t \pi_{t+1}^w \\
\mu_t^w &= w_t - (\gamma c_t + \varphi n_t)
\end{aligned}$$

After eliminating the nominal rate r_t^N , Table 2 reports the extended first-order representation. Relative to the baseline system, the table uses the dividend gap $d_t \equiv (D_t - D)/Y$, includes the government-spending deviation $g_t \equiv (G_t - G)/Y$, and keeps the steady-state labor share fixed at $1 - \alpha$.

Table 2: Extended first-order representation

No.	Name	Equation
1	Type S Euler equation	$r_t = \gamma \mathbb{E}_t \Delta c_{t+1}^S$
2	Type H budget constraint	$c_t^H = \frac{1-\alpha}{1-\Gamma}(w_t^H + n_t^H) + \frac{1}{1-\Gamma}(z_t - g_t)$
3	Aggregate consumption aggregation identity	$c_t = \lambda c_t^H + (1 - \lambda)c_t^S$
4	Aggregate labor aggregation identity	$n_t = \lambda n_t^H + (1 - \lambda)n_t^S$
5	Aggregate wage Phillips curve	$\pi_t^w = -(1 - \alpha)(\psi_w/\eta_w)\mu_t^w + \beta \mathbb{E}_t \pi_{t+1}^w$
6	Aggregate wage-markup definition	$\mu_t^w = w_t - \gamma c_t - \varphi n_t$
7	Aggregate wage-inflation identity	$\pi_t^w = w_t - w_{t-1} + \pi_t^p$
8	Type H wage Phillips curve	$\pi_t^H = -(1 - \alpha)(\psi_w/\eta_w)\mu_t^H + \beta \mathbb{E}_t \pi_{t+1}^H$
9	Type H wage-markup definition	$\mu_t^H = w_t^H - \gamma c_t^H - \varphi n_t^H$
10	Type H wage-inflation identity	$\pi_t^H = w_t^H - w_{t-1}^H + \pi_t^p$
11	Type H labor demand	$n_t^H = n_t - \psi_w(w_t^H - w_t)$
12	Aggregate wage aggregation identity	$w_t = \lambda w_t^H + (1 - \lambda)w_t^S$
13	Aggregate wage-markup aggregation identity	$\mu_t^w = \lambda \mu_t^H + (1 - \lambda)\mu_t^S$
14	Aggregate wage-inflation aggregation identity	$\pi_t^w = \lambda \pi_t^H + (1 - \lambda)\pi_t^S$
15	Price Phillips curve	$\pi_t^p = -(\psi_p/\eta_p)\mu_t^p + \beta \mathbb{E}_t \pi_{t+1}^p$
16	Price-markup definition	$\mu_t^p = y_t - n_t - w_t$
17	Dividend gap	$d_t = y_t - (1 - \alpha)(w_t + n_t)$
18	Resource constraint	$y_t = (1 - \Gamma)c_t + g_t$
19	Production function	$y_t = a_t + (1 - \alpha)n_t$
20	Taylor rule	$r_t = \phi \pi_t^p - \mathbb{E}_t \pi_{t+1}^p - m_t$

For convenience, Table 3 lists the endogenous variables appearing in Table 2, and Table 4 reports the core structural parameters used below.

Table 3: Endogenous variables in the nonredundant system

No.	Meaning	Notation
1	Output	y_t
2	Aggregate consumption	c_t
3	Aggregate labor	n_t
4	Aggregate real wage	w_t
5	Dividend gap	d_t
6	Ex ante real interest rate	r_t
7	Price markup	μ_t^p
8	Price inflation	π_t^p
9	Aggregate wage markup	μ_t^w
10	Aggregate wage inflation	π_t^w
11	Type H consumption	c_t^H
12	Type S consumption	c_t^S
13	Type H labor	n_t^H
14	Type S labor	n_t^S
15	Type H real wage	w_t^H
16	Type S real wage	w_t^S
17	Type H wage markup	μ_t^H
18	Type S wage markup	μ_t^S
19	Type H wage inflation	π_t^H
20	Type S wage inflation	π_t^S

Table 4: Core structural parameters

No.	Meaning	Parameter
1	Inverse of elasticity of intertemporal substitution	$\gamma > 0$
2	Inverse of Frisch elasticity	$\varphi > 0$
3	Discount factor	$0 < \beta < 1$
4	Elasticity of substitution across intermediate-goods varieties	$\psi_p \geq 1$
5	Price adjustment cost	$\eta_p \geq 0$
6	Elasticity of substitution across labor varieties	$\psi_w \geq 1$
7	Wage adjustment cost	$\eta_w \geq 0$
8	Taylor-rule coefficient on price inflation	$\phi > 1$
9	Share of Type H households	$0 \leq \lambda < 1$
10	Production curvature	$0 \leq \alpha < 1$
11	Steady-state government-spending share	$0 \leq \Gamma < 1$